Lab 1

Bài 1: Minh chứng kết quả

```
hadoop/lab1/output_bai1
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai1
hdfs dfs -cat hadoop/lab1/output_bai1/part-r-00000
Found 2 items
-rw-r--r--   1 bvan2925 supergroup          0 2026-03-30 15:17 hadoop/lab1/output_bai1/_SUCCESS
-rw-r--r--   1 bvan2925 supergroup       3185 2026-03-30 15:17 hadoop/lab1/output_bai1/part-r-00000
American Beauty (1999)  Average rating: 3.75 (Total ratings: 2)
Avatar (2009)   Average rating: 4.75 (Total ratings: 2)
Back to the Future (1985)       Average rating: 4.00 (Total ratings: 2)
Birdman (2014)  Average rating: 3.00 (Total ratings: 2)
Braveheart (1995)       Average rating: 4.25 (Total ratings: 2)
Coco (2017)     Average rating: 4.25 (Total ratings: 2)
Dunkirk (2017)  Average rating: 3.50 (Total ratings: 2)
Fight Club (1999)       Average rating: 4.25 (Total ratings: 2)
Forrest Gump (1994)     Average rating: 3.75 (Total ratings: 2)
Gladiator (2000)        Average rating: 3.25 (Total ratings: 2)
Good Will Hunting (1997)        Average rating: 3.50 (Total ratings: 2)
Gravity (2013)  Average rating: 4.25 (Total ratings: 2)
Inception (2010)        Average rating: 3.75 (Total ratings: 2)
Interstellar (2014)     Average rating: 3.50 (Total ratings: 2)
Jurassic Park (1993)    Average rating: 4.00 (Total ratings: 2)
La La Land (2016)       Average rating: 4.00 (Total ratings: 2)
Mad Max: Fury Road (2015)       Average rating: 4.00 (Total ratings: 2)
Manchester by the Sea (2016)    Average rating: 4.00 (Total ratings: 2)
Memento (2000)  Average rating: 4.00 (Total ratings: 2)
Moonlight (2016)        Average rating: 3.50 (Total ratings: 2)
Parasite (2019) Average rating: 4.00 (Total ratings: 2)
Pulp Fiction (1994)     Average rating: 4.00 (Total ratings: 2)
Saving Private Ryan (1998)      Average rating: 4.00 (Total ratings: 2)
Schindler's List (1993) Average rating: 4.25 (Total ratings: 2)
Se7en (1995)    Average rating: 3.75 (Total ratings: 2)
Slumdog Millionaire (2008)      Average rating: 4.25 (Total ratings: 2)
Spotlight (2015)        Average rating: 4.25 (Total ratings: 2)
Star Wars: A New Hope (1977)    Average rating: 3.50 (Total ratings: 2)
The Avengers (2012)     Average rating: 3.75 (Total ratings: 2)
The Big Short (2015)    Average rating: 4.50 (Total ratings: 2)
The Dark Knight (2008)  Average rating: 3.50 (Total ratings: 2)
The Departed (2006)     Average rating: 3.25 (Total ratings: 2)
The Godfather (1972)    Average rating: 3.50 (Total ratings: 2)
```

```
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai1
hdfs dfs -cat hadoop/lab1/output_bai1/part-r-00000
Jurassic Park (1993)    Average rating: 4.00 (Total ratings: 2)
La La Land (2016)       Average rating: 4.00 (Total ratings: 2)
Mad Max: Fury Road (2015)       Average rating: 4.00 (Total ratings: 2)
Manchester by the Sea (2016)    Average rating: 4.00 (Total ratings: 2)
Memento (2000)  Average rating: 4.00 (Total ratings: 2)
Moonlight (2016)        Average rating: 3.50 (Total ratings: 2)
Parasite (2019) Average rating: 4.00 (Total ratings: 2)
Pulp Fiction (1994)     Average rating: 4.00 (Total ratings: 2)
Saving Private Ryan (1998)      Average rating: 4.00 (Total ratings: 2)
Schindler's List (1993) Average rating: 4.25 (Total ratings: 2)
Se7en (1995)    Average rating: 3.75 (Total ratings: 2)
Slumdog Millionaire (2008)      Average rating: 4.25 (Total ratings: 2)
Spotlight (2015)        Average rating: 4.25 (Total ratings: 2)
Star Wars: A New Hope (1977)    Average rating: 3.50 (Total ratings: 2)
The Avengers (2012)     Average rating: 3.75 (Total ratings: 2)
The Big Short (2015)    Average rating: 4.50 (Total ratings: 2)
The Dark Knight (2008)  Average rating: 3.50 (Total ratings: 2)
The Departed (2006)     Average rating: 3.25 (Total ratings: 2)
The Godfather (1972)    Average rating: 3.50 (Total ratings: 2)
The Green Mile (1999)   Average rating: 3.50 (Total ratings: 2)
The Imitation Game (2014)       Average rating: 3.75 (Total ratings: 2)
The Lion King (1994)    Average rating: 4.00 (Total ratings: 2)
The Lord of the Rings: The Fellowship of the Ring (2001)        Average rating: 4.25 (Total ratings: 2)
The Lord of the Rings: The Return of the King (2003)    Average rating: 3.50 (Total ratings: 2)
The Lord of the Rings: The Two Towers (2002)    Average rating: 3.75 (Total ratings: 2)
The Matrix (1999)       Average rating: 3.75 (Total ratings: 2)
The Prestige (2006)     Average rating: 4.00 (Total ratings: 2)
The Revenant (2015)     Average rating: 3.50 (Total ratings: 2)
The Shawshank Redemption (1994) Average rating: 4.00 (Total ratings: 2)
The Silence of the Lambs (1991) Average rating: 3.50 (Total ratings: 2)
The Social Network (2010)       Average rating: 3.75 (Total ratings: 2)
The Usual Suspects (1995)       Average rating: 4.25 (Total ratings: 2)
The Wolf of Wall Street (2013)  Average rating: 4.25 (Total ratings: 2)
Titanic (1997)  Average rating: 3.50 (Total ratings: 2)
Toy Story (1995)        Average rating: 4.25 (Total ratings: 2)
Whiplash (2014) Average rating: 3.75 (Total ratings: 2)

Notice: No movie found with at least 5 ratings.
bvan2925@MSI:~/hadoop$
```

Bài 2: Minh chứng kết quả

```
bvan2925@MSI:~/hadoop$ hadoop jar bai2.jar bai2 \
hadoop/lab1/input/movies.txt \
hadoop/lab1/input/ratings_*.txt \
hadoop/lab1/output_bai2
                Merged Map outputs=3
                GC time elapsed (ms)=29
                Total committed heap usage (bytes)=847249408
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=479
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai2
hdfs dfs -cat hadoop/lab1/output_bai2/part-r-00000
Found 2 items
-rw-r--r--   1 bvan2925 supergroup          0 2026-03-30 14:38 hadoop/lab1/output_bai2/_SUCCESS
-rw-r--r--   1 bvan2925 supergroup        479 2026-03-30 14:38 hadoop/lab1/output_bai2/part-r-00000
Action  Avg: 3.73, Count: 20
Adventure       Avg: 3.85, Count: 30
Animation       Avg: 4.17, Count: 6
Biography       Avg: 4.14, Count: 14
Children        Avg: 4.25, Count: 2
Comedy  Avg: 3.96, Count: 12
Crime   Avg: 3.78, Count: 20
Drama   Avg: 3.81, Count: 76
Family  Avg: 4.25, Count: 2
Fantasy Avg: 3.95, Count: 10
History Avg: 4.00, Count: 6
Music   Avg: 3.88, Count: 4
Mystery Avg: 4.00, Count: 8
Romance Avg: 3.75, Count: 8
Sci-Fi  Avg: 3.85, Count: 20
Thriller        Avg: 3.88, Count: 16
War     Avg: 4.00, Count: 2
bvan2925@MSI:~/hadoop$
```

Bài 3: Minh chứng kết quả

```
hadoop/lab1/output_bai3
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai3
hdfs dfs -cat hadoop/lab1/output_bai3/part-r-00000
Found 2 items
-rw-r--r--   1 bvan2925 supergroup          0 2026-03-30 14:41 hadoop/lab1/output_bai3/_SUCCESS
-rw-r--r--   1 bvan2925 supergroup       2386 2026-03-30 14:41 hadoop/lab1/output_bai3/part-r-00000
American Beauty (1999)  Male: 3.00, Female: 4.50
Avatar (2009)   Male: 5.00, Female: 4.50
Back to the Future (1985)       Male: 4.00, Female: 4.00
Birdman (2014)  Male: 3.00, Female: 3.00
Braveheart (1995)       Male: 4.50, Female: 4.00
Coco (2017)     Male: 5.00, Female: 3.50
Dunkirk (2017)  Male: 4.00, Female: 3.00
Fight Club (1999)       Male: 5.00, Female: 3.50
Forrest Gump (1994)     Male: 2.50, Female: 5.00
Gladiator (2000)        Male: 3.50, Female: 3.00
Good Will Hunting (1997)        Male: 3.00, Female: 4.00
Gravity (2013)  Male: 4.00, Female: 4.50
Inception (2010)        Male: 3.50, Female: 4.00
Interstellar (2014)     Male: 3.50, Female: 3.50
Jurassic Park (1993)    Male: 3.50, Female: 4.50
La La Land (2016)       Male: 5.00, Female: 3.00
Mad Max: Fury Road (2015)       Male: 4.00, Female: 4.00
Manchester by the Sea (2016)    Male: 3.50, Female: 4.50
Memento (2000)  Male: 3.50, Female: 4.50
Moonlight (2016)        Male: 3.00, Female: 4.00
Parasite (2019) Male: 4.00, Female: 4.00
Pulp Fiction (1994)     Male: 4.50, Female: 3.50
Saving Private Ryan (1998)      Male: 4.00, Female: 4.00
Schindler's List (1993) Male: 4.00, Female: 4.50
Se7en (1995)    Male: 4.00, Female: 3.50
Slumdog Millionaire (2008)      Male: 4.50, Female: 4.00
Spotlight (2015)        Male: 4.50, Female: 4.00
Star Wars: A New Hope (1977)    Male: 4.00, Female: 3.00
The Avengers (2012)     Male: 3.50, Female: 4.00
The Big Short (2015)    Male: 5.00, Female: 4.00
The Dark Knight (2008)  Male: 4.00, Female: 3.00
The Departed (2006)     Male: 4.50, Female: 2.00
The Godfather (1972)    Male: 4.50, Female: 2.50
The Green Mile (1999)   Male: 3.00, Female: 4.00
The Imitation Game (2014)       Male: 4.50, Female: 3.00
```

```
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai3
hdfs dfs -cat hadoop/lab1/output_bai3/part-r-00000
Inception (2010)        Male: 3.50, Female: 4.00
Interstellar (2014)     Male: 3.50, Female: 3.50
Jurassic Park (1993)    Male: 3.50, Female: 4.50
La La Land (2016)       Male: 5.00, Female: 3.00
Mad Max: Fury Road (2015)       Male: 4.00, Female: 4.00
Manchester by the Sea (2016)    Male: 3.50, Female: 4.50
Memento (2000)  Male: 3.50, Female: 4.50
Moonlight (2016)        Male: 3.00, Female: 4.00
Parasite (2019) Male: 4.00, Female: 4.00
Pulp Fiction (1994)     Male: 4.50, Female: 3.50
Saving Private Ryan (1998)      Male: 4.00, Female: 4.00
Schindler's List (1993) Male: 4.00, Female: 4.50
Se7en (1995)    Male: 4.00, Female: 3.50
Slumdog Millionaire (2008)      Male: 4.50, Female: 4.00
Spotlight (2015)        Male: 4.50, Female: 4.00
Star Wars: A New Hope (1977)    Male: 4.00, Female: 3.00
The Avengers (2012)     Male: 3.50, Female: 4.00
The Big Short (2015)    Male: 5.00, Female: 4.00
The Dark Knight (2008)  Male: 4.00, Female: 3.00
The Departed (2006)     Male: 4.50, Female: 2.00
The Godfather (1972)    Male: 4.50, Female: 2.50
The Green Mile (1999)   Male: 3.00, Female: 4.00
The Imitation Game (2014)       Male: 4.50, Female: 3.00
The Lion King (1994)    Male: 4.50, Female: 3.50
The Lord of the Rings: The Fellowship of the Ring (2001)        Male: 4.50, Female: 4.00
The Lord of the Rings: The Return of the King (2003)    Male: 4.00, Female: 3.00
The Lord of the Rings: The Two Towers (2002)    Male: 4.50, Female: 3.00
The Matrix (1999)       Male: 3.00, Female: 4.50
The Prestige (2006)     Male: 4.00, Female: 4.00
The Revenant (2015)     Male: 3.50, Female: 3.50
The Shawshank Redemption (1994) Male: 3.00, Female: 5.00
The Silence of the Lambs (1991) Male: 3.50, Female: 3.50
The Social Network (2010)       Male: 3.50, Female: 4.00
The Usual Suspects (1995)       Male: 4.00, Female: 4.50
The Wolf of Wall Street (2013)  Male: 4.00, Female: 4.50
Titanic (1997)  Male: 4.00, Female: 3.00
Toy Story (1995)        Male: 4.50, Female: 4.00
Whiplash (2014) Male: 4.50, Female: 3.00
bvan2925@MSI:~/hadoop$
```

Bài 4: Minh chứng kết quả

```
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai4
hdfs dfs -cat hadoop/lab1/output_bai4/part-r-00000
-rw-r--r--   1 bvan2925 supergroup          0 2026-03-30 14:43 hadoop/lab1/output_bai4/_SUCCESS
-rw-r--r--   1 bvan2925 supergroup       3442 2026-03-30 14:43 hadoop/lab1/output_bai4/part-r-00000
American Beauty (1999)  0-18: NA   18-35: 3.75   35-50: NA   50+: NA
Avatar (2009)   0-18: NA   18-35: 4.50   35-50: 5.00   50+: NA
Back to the Future (1985)       0-18: NA   18-35: 4.00   35-50: NA   50+: NA
Birdman (2014)  0-18: NA   18-35: 3.00   35-50: NA   50+: NA
Braveheart (1995)       0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
Coco (2017)     0-18: NA   18-35: 4.25   35-50: NA   50+: NA
Dunkirk (2017)  0-18: NA   18-35: 3.50   35-50: NA   50+: NA
Fight Club (1999)       0-18: NA   18-35: 4.25   35-50: NA   50+: NA
Forrest Gump (1994)     0-18: NA   18-35: 3.75   35-50: NA   50+: NA
Gladiator (2000)        0-18: NA   18-35: 3.00   35-50: 3.50   50+: NA
Good Will Hunting (1997)        0-18: NA   18-35: 3.50   35-50: NA   50+: NA
Gravity (2013)  0-18: NA   18-35: 4.50   35-50: 4.00   50+: NA
Inception (2010)        0-18: NA   18-35: 3.75   35-50: NA   50+: NA
Interstellar (2014)     0-18: NA   18-35: 3.50   35-50: 3.50   50+: NA
Jurassic Park (1993)    0-18: NA   18-35: 4.00   35-50: NA   50+: NA
La La Land (2016)       0-18: NA   18-35: 3.00   35-50: 5.00   50+: NA
Mad Max: Fury Road (2015)       0-18: NA   18-35: 4.00   35-50: NA   50+: NA
Manchester by the Sea (2016)    0-18: NA   18-35: 4.50   35-50: 3.50   50+: NA
Memento (2000)  0-18: NA   18-35: 4.50   35-50: 3.50   50+: NA
Moonlight (2016)        0-18: NA   18-35: 4.00   35-50: 3.00   50+: NA
Parasite (2019) 0-18: NA   18-35: 4.00   35-50: 4.00   50+: NA
Pulp Fiction (1994)     0-18: NA   18-35: 4.00   35-50: NA   50+: NA
Saving Private Ryan (1998)      0-18: NA   18-35: 4.00   35-50: 4.00   50+: NA
Schindler's List (1993) 0-18: NA   18-35: 4.50   35-50: 4.00   50+: NA
Se7en (1995)    0-18: NA   18-35: 3.75   35-50: NA   50+: NA
Slumdog Millionaire (2008)      0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
Spotlight (2015)        0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
Star Wars: A New Hope (1977)    0-18: NA   18-35: 3.00   35-50: 4.00   50+: NA
The Avengers (2012)     0-18: NA   18-35: 3.75   35-50: NA   50+: NA
The Big Short (2015)    0-18: NA   18-35: 4.50   35-50: NA   50+: NA
The Dark Knight (2008)  0-18: NA   18-35: 3.00   35-50: 4.00   50+: NA
The Departed (2006)     0-18: NA   18-35: 2.00   35-50: 4.50   50+: NA
The Godfather (1972)    0-18: NA   18-35: 3.50   35-50: NA   50+: NA
The Green Mile (1999)   0-18: NA   18-35: 4.00   35-50: 3.00   50+: NA
The Imitation Game (2014)       0-18: NA   18-35: 3.00   35-50: 4.50   50+: NA
The Lion King (1994)    0-18: NA   18-35: 4.00   35-50: NA   50+: NA
The Lord of the Rings: The Fellowship of the Ring (2001)        0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
```

```
bvan2925@MSI:~/hadoop$ hdfs dfs -ls hadoop/lab1/output_bai4
hdfs dfs -cat hadoop/lab1/output_bai4/part-r-00000
Inception (2010)        0-18: NA   18-35: 3.75   35-50: NA   50+: NA
Interstellar (2014)     0-18: NA   18-35: 3.50   35-50: 3.50   50+: NA
Jurassic Park (1993)    0-18: NA   18-35: 4.00   35-50: NA   50+: NA
La La Land (2016)       0-18: NA   18-35: 3.00   35-50: 5.00   50+: NA
Mad Max: Fury Road (2015)       0-18: NA   18-35: 4.00   35-50: NA   50+: NA
Manchester by the Sea (2016)    0-18: NA   18-35: 4.50   35-50: 3.50   50+: NA
Memento (2000)  0-18: NA   18-35: 4.50   35-50: 3.50   50+: NA
Moonlight (2016)        0-18: NA   18-35: 4.00   35-50: 3.00   50+: NA
Parasite (2019) 0-18: NA   18-35: 4.00   35-50: 4.00   50+: NA
Pulp Fiction (1994)     0-18: NA   18-35: 4.00   35-50: NA   50+: NA
Saving Private Ryan (1998)      0-18: NA   18-35: 4.00   35-50: 4.00   50+: NA
Schindler's List (1993) 0-18: NA   18-35: 4.50   35-50: 4.00   50+: NA
Se7en (1995)    0-18: NA   18-35: 3.75   35-50: NA   50+: NA
Slumdog Millionaire (2008)      0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
Spotlight (2015)        0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
Star Wars: A New Hope (1977)    0-18: NA   18-35: 3.00   35-50: 4.00   50+: NA
The Avengers (2012)     0-18: NA   18-35: 3.75   35-50: NA   50+: NA
The Big Short (2015)    0-18: NA   18-35: 4.50   35-50: NA   50+: NA
The Dark Knight (2008)  0-18: NA   18-35: 3.00   35-50: 4.00   50+: NA
The Departed (2006)     0-18: NA   18-35: 2.00   35-50: 4.50   50+: NA
The Godfather (1972)    0-18: NA   18-35: 3.50   35-50: NA   50+: NA
The Green Mile (1999)   0-18: NA   18-35: 4.00   35-50: 3.00   50+: NA
The Imitation Game (2014)       0-18: NA   18-35: 3.00   35-50: 4.50   50+: NA
The Lion King (1994)    0-18: NA   18-35: 4.00   35-50: NA   50+: NA
The Lord of the Rings: The Fellowship of the Ring (2001)        0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
The Lord of the Rings: The Return of the King (2003)    0-18: NA   18-35: 3.50   35-50: NA   50+: NA
The Lord of the Rings: The Two Towers (2002)    0-18: NA   18-35: 3.00   35-50: 4.50   50+: NA
The Matrix (1999)       0-18: NA   18-35: 3.75   35-50: NA   50+: NA
The Prestige (2006)     0-18: NA   18-35: 4.00   35-50: 4.00   50+: NA
The Revenant (2015)     0-18: NA   18-35: 3.50   35-50: NA   50+: NA
The Shawshank Redemption (1994) 0-18: NA   18-35: 4.00   35-50: NA   50+: NA
The Silence of the Lambs (1991) 0-18: NA   18-35: 3.50   35-50: 3.50   50+: NA
The Social Network (2010)       0-18: NA   18-35: 4.00   35-50: 3.50   50+: NA
The Usual Suspects (1995)       0-18: NA   18-35: 4.25   35-50: NA   50+: NA
The Wolf of Wall Street (2013)  0-18: NA   18-35: 4.50   35-50: 4.00   50+: NA
Titanic (1997)  0-18: NA   18-35: 3.00   35-50: 4.00   50+: NA
Toy Story (1995)        0-18: NA   18-35: 4.00   35-50: 4.50   50+: NA
Whiplash (2014) 0-18: NA   18-35: 3.00   35-50: 4.50   50+: NA
bvan2925@MSI:~/hadoop$
```